Antonio Martin-Ozimek

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Education

The University of Alberta – Bilingual BSc in Computer Engineering, Co-op Class of 2025
Key Classes – Object Oriented, Tangible Computing, Computer Logic, Microprocessors, Databases
Skills – Python, C++, MATLAB, Arduino, HTML5, CSS, C, JavaScript, Unity Engine, C#, OpenCV, React, Tensorflow, NextJS, Keras, Google Colab, SQL, YOLOv8, ESP32, Embedded Systems Programming

Experience

University of Waterloo – ML Researcher May 2025 – August 2025

- Analyzed three datasets using **Python** (**pandas, NumPy, scikit-learn**) to model stress-**pHRI** relationships with LSTM neural networks (**PyTorch**)
- Led **pilot study** based on previous findings to address data limitations and enhance stress-**pHRI** research protocols

HONDA Research Institute - ML Researcher May 2024 – December 2024

- First author on two short papers on generating human-like behaviours to **HRI2025**
- Implemented two **generative models**: an **Implicit Behavior Cloning** and a **Diffusion Behavior Cloning** model to generate non-verbal cues from **360-degree images** using **Tensorflow** and **PyTorch**
- Extracted **feature** data using **YOLOPose** and **Gaze360** to then be filtered using **pandas** for thresholding, normalization, and interpolation
- Tracked training results using **Tensorboard** and **WandB** for **MLOps** backbone

Enterprises Macay Inc. - ML Engineer May 2023 – September 2023

- Headed the implementation of a **CNN model** to synchronously detect moisture level in **infrared images** and catch contaminants in **RGB** images in quality control line
- Designed and implemented two models: a pre-trained **YOLO-v8 CNN** model fine-tuned on **IR images** and a **CNN** using **Keras** and **TensorFlow** trained from scratch on in-house dataset

University of Alberta - ML Researcher January 2023 – April 2023

- Engineered a sensor fusion **PID controller** integrating **YOLOv8 real-time object tracking** with ultrasonic distance sensors for setpoint maintenance in a MIMO system
- Architected **OPCUA client-server** communication system bridging embedded controllers with legacy Windows 7 systems, delivering $5,000 cost savings by eliminating hardware upgrade requirements
- Developed machine learning pipeline using **PySindy** and **Optuna** for automated **system modelling** and **parameter tuning**

Projects

YoloBrawlers, HackED2025 (OVERALL WINNER) | Python, C++, Computer Vision, Pose Detection, ESP32

- Engineered dual autonomous boxing robots with servo-controlled weaving and punching mechanisms, **winning over $3,000 in prize money** through innovative robotic combat design
- Developed real-time **computer vision control system** using **YOLOv11 for pose detection** and **behavioral translation**, creating custom signal schemes
- Optimized **ESP32 microcontroller** implementation to ensure real-time, reliable robot

60g BattleBot, WATBOT2025 (SECOND PLACE) | FreeCAD, C/C++

- Designed and iteratively optimized 60g combat robot using **FreeCAD**, engineering weapon systems, chassis, and protective shell to compete against 21 other teams
- Fabricated **dual parallel circuit** architecture with precision soldering to ensure reliable **ESC**-to-motor control while maintaining integrated safety switch functionality

Autonomous Driving Robot | Python, Docker, Duckietown, ROS2

- Developing an **intelligent agent** for an **autonomous driving** robot to complete **lane-following** and **localization** tasks using **ROS2** to maintain system